Flame test notes

What the test shows

Some metal compounds produce characteristic colours in a flame. Heat excites electrons in gaseous species. Light is emitted as electrons return to lower energy levels.

Observation table

Sodium (Na): golden yellow. Calcium (Ca): brick red. Barium (Ba): apple green.

Interpreting a result

Compare an unknown with reference colours. Traces of sodium may mask weaker colours. A flame test is a preliminary identification and may require a confirmatory test.

Try it online

Open Qualitative Chemistry in the student portal and select the Flame test. Compare all three observations before taking the MCQ quiz.